

Settled Weight: A Sealed, Zero-Parameter Test of the Solvency Field Theory

Weak-Field Law on SPARC, with a Registered Falsification and Its Repair

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Abstract

We report a sealed, dual-implemented test of the Solvency Field Theory (SFT) weak-field law on the 175 late-type galaxies of the SPARC database. The law, $g^2 = g_{\text{bar}}^2 + \lambda_{\text{eff}} a_S g_{\text{bar}}$ with derived scale $a_S = cH_0/2\pi$, was evaluated with *zero free parameters*: all coefficients are geometric constants or cited literature values, and all predictions were generated from baryonic source columns only, hash-sealed before the observed velocities were unsealed. Three results follow. (1) The data’s preferred effective deep-regime scale, $\lambda^* a_S = 1.211 \times 10^{-10} \text{ m s}^{-2}$, lies within 0.9% of the empirical radial-acceleration-relation scale, quantitatively realizing the long-noted $a_0 \sim cH_0$ coincidence with a derived scale and an $O(1)$ geometric coefficient. (2) A geometry-only coefficient model reached 0.45% parity with the best globally fitted constant λ^* , won four predeclared high-quality subgroups, and was *falsified* in its registered gas-rich dwarf cell. (3) A successor operator—participation-weighted settlement, which discounts the coefficient of dissipative gas by its coherent support fraction—was frozen from theory before any further target contact and *defeats the fitted constant on the full sample* (0.09730 vs. 0.09865 dex in per-point $\log_{10} V$ RMS), helping 57% of galaxies individually, while leaving a quantified residual in the most gas-dominated dwarf cell and a marginal parity breach in the highest-quality subgroup, both reported. The falsification, the repair, and the residuals were all obtained under a pre-registered protocol with dual independent implementations agreeing at machine precision; every number in this paper reproduces from hashed public artifacts. The operator’s central claim—that the weak-field boost is sourced by accumulated coherent settlement history rather than instantaneous mass shape—predicts systematically weaker boosts in kinematically unsettled systems, including a rise of λ_{eff} with cosmic time testable against high-redshift rotation curves.

1 Introduction

Rotation curves of disk galaxies depart from Newtonian expectations below a characteristic acceleration $a_0 \approx 1.2 \times 10^{-10} \text{ m s}^{-2}$, a regularity sharpened by the radial acceleration relation (RAR) (McGaugh et al., 2016) on the SPARC sample (Lelli et al., 2016). Milgrom noted at MOND’s founding that a_0 is numerically close to $cH_0/2\pi$ (Milgrom, 1983); the coincidence has remained underived in the phenomenology (Famaey & McGaugh, 2012), and emergent-gravity approaches place related scales at $cH_0/6$ (Verlinde, 2017).

Solvency Field Theory (SFT) is an independent research program in which spacetime structure is the accumulated record of discrete, irreversible “settlement” events sourced by matter; its derivations

are archived with reproduction packages (Fosmark, 2026). In its weak-field sector SFT predicts the acceleration scale $a_S = cH_0/2\pi$ outright and predicts that the $O(1)$ coefficient λ multiplying the anomalous term is not universal but set by source structure, bounded by two parameter-free geometric endpoints: $\lambda_{\text{sph}} = 4/3$ for isotropic support and $\lambda_{\text{disk}} = 3\pi/8$ for planar support.

This paper reports a complete, sealed test of that claim—including the sealed falsification of one of its registered predictions and the frozen, no-tuning repair that followed. We consider the conduct of the test to be part of its content: an earlier analysis by this program (Sec. 3) found an apparent morphology–coefficient correlation that our own internal audit later showed to be carried by a systematics-dominated subsample. The present work replaces that analysis with a forward-modeling protocol in which predictions were generated from baryonic source data only, dual-implemented by two independent computational engines with no shared code, hash-sealed, and only then compared with observed velocities under pre-registered metrics, controls, subgroups, and verdict criteria.

2 The law, the scale, and the geometric band

The SFT weak-field law tested here is

$$g_{\text{SFT}}^2(r) = g_{\text{bar}}^2(r) + \lambda_{\text{eff}}(r) a_S g_{\text{bar}}(r), \quad a_S = \frac{cH_0}{2\pi}, \quad (1)$$

equivalent to the interpolating family $\nu(y) = \sqrt{1 + \lambda/y}$. With $H_0 = 67.74 \text{ km s}^{-1} \text{ Mpc}^{-1}$ (Planck Collaboration, 2020), $a_S = 1.0475 \times 10^{-10} \text{ m s}^{-2}$; H_0 is the only imported cosmological quantity and is not adjusted. The coefficient endpoints $4/3$ and $3\pi/8$ arise, respectively, from isotropic and planar projection of source maintenance structure onto the circular-velocity observable; their derivations, including the admissibility factor $3/4$ entering the disk endpoint, are in the archived theory documents (Fosmark, 2026). No quantity in Eq. (1) is fitted anywhere in this paper. Provenance of the endpoints relative to the data analyses is documented by the archive’s version history and by hashed sealed-prediction records included in the reproduction package.

3 Why a sealed retest: the autopsy of a fitted correlation

An earlier analysis in this program profiled a per-galaxy best-fit λ against SPARC rotation curves and reported a morphology correlation (Spearman $\rho = -0.27$, $p = 2.8 \times 10^{-4}$) consistent in sign with the geometric band. Our internal re-audit, conducted before submission, found: (i) the correlation is carried by the $T \geq 10$ dwarf-irregular subsample and vanishes within quality-controlled spirals ($\rho \approx 0$); (ii) only 6.3% of fitted λ values fall inside the predicted band, with fits railing at both scan boundaries; (iii) the fitted per-galaxy λ behaves as a nuisance absorber for mass-to-light, distance, and functional-form error rather than as a geometric measurement. The fitted- λ observable was therefore retired. The full autopsy, including the original outputs, is preserved in the reproduction package as the motivation for the forward protocol below; the earlier record is superseded, not deleted.

4 Protocol

Data classes. Inputs were partitioned before analysis into *source-only* (radius; gas, disk, and bulge velocity components; surface-brightness profiles), *target-locked* (V_{obs} , eV_{obs} , and all quantities derived from them, including the public RAR tables), and *shared infrastructure* (sample metadata: Hubble type T , quality flag Q), with provenance flags recorded.

Model. Predictions use Eq. (1) with $\lambda_{\text{eff}}(r) = \sum_j w_j(r) p_j(r) \lambda_j$, where j runs over bulge, stellar disk, and gas; w_j is the component’s nonnegative share of the local baryonic support, $S_{\text{disk}} = \Upsilon_d V_{\text{disk}}^2$, $S_{\text{bul}} = \Upsilon_b V_{\text{bul}}^2$, $S_{\text{gas}} = |V_{\text{gas}}|^2$; the signed ledger $V_{\text{bar}}^2 = \Upsilon_d V_{\text{disk}}^2 + \Upsilon_b V_{\text{bul}}^2 + V_{\text{gas}} |V_{\text{gas}}|$ sources g_{bar} ; $\Upsilon_d = 0.5$, $\Upsilon_b = 0.7$ at $3.6 \mu\text{m}$ (Schombert et al., 2019). Component coefficients: $\lambda_{\text{bul}} = 4/3$; $\lambda_{\text{star}} = 4/3 - (4/3 - 3\pi/8)(1 - h_z/R_d)$ with $R_d/h_z = 7.3$ (Kregel et al., 2002), giving $\lambda_{\text{star}} = 1.1994$; $\lambda_{\text{gas}} = 4/3$. The geometry-only model sets all participation factors $p_j = 1$. The participation-weighted model (Sec. 5.4) sets $p_{\text{star}} = p_{\text{bul}} = 1$ (collisionless, recurrent) and

$$p_{\text{gas}}(r) = \frac{V_{\text{src}}^2(r)}{V_{\text{src}}^2(r) + 3\sigma_{1\text{D}}^2}, \quad V_{\text{src}}^2 = \max(V_{\text{bar}}^2, 0), \quad \sigma_{1\text{D}} = 10 \text{ km s}^{-1}, \quad (2)$$

the ordered-to-total kinetic support ratio of dissipative gas; the factor 3 converts the measured line-of-sight dispersion (Leroy et al., 2008) to total turbulent support under isotropy and is geometric, not tuned.

Sealing and duality. Two independent engines, built from a common frozen specification with no shared code, generated all predictions from source-only columns; the two sets agree to $\leq 5 \times 10^{-7} \text{ km s}^{-1}$ at every one of 3,391 radial points, with the identical two rows flagged by the declared nonpositive-source rule. Predictions were hashed before the target columns were read. Validation metrics, controls, row rules, eleven subgroup cells, parity threshold, and PASS/PARTIAL/FAIL criteria were frozen in writing before unsealing; scoring was likewise dual-implemented (agreement to ten significant figures). The participation operator of Eq. (2) was frozen—with its conventions, declared expectations, and kill criteria—before any renewed target contact, in a chartered fork whose founding document predates its predictions. One sequencing deviation (an early source-side model diagnostic by one engine) is logged in the reproduction package with severity assessment. All artifacts are archived with SHA-256 hashes (Fosmark, 2026).

5 Results

Scored rows: 3,389 (two flagged rows excluded by rule). The best globally fitted constant on these rows is $\lambda^* = 1.156$.

5.1 The acceleration scale

The fitted control carries the scale result: $\lambda^* a_S = 1.211 \times 10^{-10} \text{ m s}^{-2}$, within 0.9% ($\approx 0.5\sigma$ statistical) of the empirical $a_0 = (1.20 \pm 0.02) \times 10^{-10} \text{ m s}^{-2}$ (McGaugh et al., 2016); equivalently the data select $a_0 = cH_0/5.44$, between Milgrom’s 2π (Milgrom, 1983) and Verlinde’s 6 (Verlinde, 2017). The fitted coefficient sits 1.9% below the geometric band $[3\pi/8, 4/3]$; the band floor is recovered exactly for $H_0 = 66.5 \text{ km s}^{-1} \text{ Mpc}^{-1}$, so the offset is degenerate with the Hubble tension and the Υ normalization at its full size. The disk endpoint $\lambda = 3\pi/8$ is the best-performing fixed control in the study.

5.2 Full-sample metrics

Table 1 reports the per-point RMS of $\log_{10} V_{\text{obs}} - \log_{10} V_{\text{pred}}$ (metric M1, primary) for all models and controls.

5.3 The geometry-only model: parity, subgroup wins, and a sealed falsification

Against λ^* the geometry-only model achieves parity (1.0045, inside the pre-registered 1.02 band) and wins four of eleven predeclared cells: $T \leq 3$, $T4-7$, $Q = 1$, and bulge-present (per-galaxy help

Model / control	M1 (dex)	Notes
Newtonian (baryons only)	0.26534	
$\lambda = 1$	0.09939	
$\lambda = 4/3$ (sphere endpoint)	0.09939	
$\lambda = 3\pi/8$ (disk endpoint)	0.09866	best fixed control
$\lambda = \text{midpoint}$ (1.2557)	0.09890	
$\lambda^* = 1.156$ (fitted, one parameter)	0.09865	target-fit rival
Geometry-only SFT (zero parameters)	0.09909	parity ratio 1.0045
Participation-weighted SFT (zero parameters)	0.09730	ratio 0.9864

Table 1: Full-sample results. Both SFT models beat Newtonian at $\sim 91\text{--}92\%$ per-galaxy help. The geometry-only model reaches 0.45% parity with the fitted constant; the participation-weighted model defeats it by 1.36%, helping 57.1% of galaxies individually.

0.54–0.62), establishing under seal that a morphology-dependent coefficient exists in the predicted direction where data quality is highest. Its registered dwarf prediction—that gas-dominated dwarfs prefer $\lambda \rightarrow 4/3$ above a frozen gas-share threshold—was *falsified in both of its predeclared cells*: gas-rich dwarfs prefer coefficients at or below λ^* . The failure is thereby localized to the gas-support sector: instantaneous shape misassigns the coefficient of matter that has not dynamically settled.

5.4 The participation-weighted repair

The successor hypothesis—that the boost is sourced by accumulated *coherent* settlement, so dissipative, decorrelating gas banks only its ordered support fraction—was frozen as Eq. (2) with declared expectations, then validated on the identical rows, metrics, controls, and cells. Outcomes: full-sample $M1 = 0.09730$, defeating λ^* (0.09865) and every fixed control; per-galaxy help vs. λ^* 57.1%; the dwarf sector flipped (T 10–11 help $0.23 \rightarrow 0.64$; low-gas dwarf cell help $0.21 \rightarrow 0.70$; $Q2$, $Q3$ to wins) while $T \leq 3$ and bulge-present wins were retained. Under the frozen validation protocol this is a **PASS**—to our knowledge the first instance in this program of a zero-parameter model outperforming a constant fitted directly to the validation data on the full SPARC sample.

Two residuals stand, reported with equal weight. (i) *Extreme-cell overshoot*: the eleven most gas-dominated dwarfs received median $\lambda_{\text{eff}} = 0.837$ where the data prefer $\sim 1.0\text{--}1.16$; the cell is lost to λ^* (help 0.455) while still marginally improving on the geometry-only model. This converts the miss into a measurement: the cell brackets a participation *floor*, effective $p \approx 0.75\text{--}0.87$ where Eq. (2) delivers $\sim 0.5\text{--}0.6$ —the first observational constraint on minimum settlement recurrence in this framework. (ii) *Quality tax*: the $Q = 1$ cell breaches the 1.02 parity band at 1.026, a $\sim 2.6\%$ degradation of the best-measured spirals attributable to the gas discount in their outskirts. Under the fork’s own stricter charter these residuals grade the repair **PARTIAL**; its predeclared kill criteria were not triggered. No parameter was adjusted at any point in response to target data; the residuals are published as physics, and any floor derivation is assigned to a future pre-registered fork.

6 Interpretation

Taken together the results support a specific mechanical reading: the weak-field anomaly’s *scale* is set by the cosmological write rate ($a_S = cH_0/2\pi$, empirically confirmed at the percent level through $\lambda^*a_S \simeq a_0$), while its *coefficient* is set by how much of the local baryonic weight has

accumulated coherent settlement history. Collisionless stars—on recurrent orbits for gigayears—bank that history fully; dissipative, turbulence-decorrelated gas banks only its coherent fraction; and systems dominated by unsettled gas exhibit weaker boosts, exactly the behavior that both the fitted- λ archaeology and the sealed forward test independently demanded. The one standing tension (the participation floor at extreme gas dominance) is a target for the theory’s settlement-relaxation operator rather than for phenomenological adjustment.

7 Registered predictions

The validated operator makes source-computable predictions beyond SPARC, registered here prior to any test. (1) *Redshift evolution*: because settlement accumulates, λ_{eff} at fixed baryonic content rises with cosmic time; high- z disks—dynamically hotter and less settled (Kassin et al., 2012)—should show systematically weaker boosts, even where JWST imaging shows early *photometric* disks (Ferreira et al., 2022): the prediction tracks kinematic settling, not apparent shape. (2) *Disturbed systems*: recent mergers and strongly interacting pairs should sit low off the RAR relative to settled twins of equal g_{bar} . (3) *Recent accretion*: newly accreted gas is boost-discounted even where dynamically cold. Existing high- z kinematic samples (KMOS^{3D}-class surveys, ALMA rotation curves) can test (1) directly.

8 Conclusion

A zero-parameter participation-weighted SFT model, born from the sealed failure of the prior geometry-only prediction, beats the best fitted global constant- λ control on the full SPARC sample while leaving a quantified residual in the most gas-dominated dwarf cell. The derived scale $cH_0/2\pi$ lands within 1% of the empirical a_0 through an $O(1)$ coefficient inside 2% of a parameter-free geometric band. We regard the sequence documented here—registered prediction, sealed falsification, frozen repair, full-sample pass, standing residuals—as inseparable from the result: every number reproduces from hashed artifacts, and the theory’s next obligation (the participation floor, and the $\lambda(z)$ prediction) is stated before the data that will judge it have been touched.

Acknowledgments

The computational protocol was executed with two independent AI engines (Anthropic Claude; OpenAI GPT-5.5) serving as canonical and adversarial-shadow implementations under the author’s refereeing; their convergences, disagreements, retractions, and logged deviations are part of the archived record. The SPARC database is due to Lelli, McGaugh & Schombert.

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